Digital Avionics — Homework 1

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Question 1

What are the advantages of digital avionics comparing the analog avionics?

Solution

Question 2

Is there any drawback of the digital avionics? How to overcome the drawback in aircraft system engineering?

Solution

Question 3

Introduce and compare the CNS/ATM and NAS (National Airspace System).

Solution

Question 4

Every aircraft equips with voice communication by radio frequency. Many radio frequencies are used onboard for different avionics. Please show all the radio frequencies and the purposes of the belonging avionics.

Solution

Question 5

Explain the relationships of Guidance, Navigation, and Control for an aircraft.

Solution

Question 6

What is the wing incidence, α, when flying straight and level at a speed of 80 m/s (160 knots approx.) at a height of 200 ft in the approach to the airfield?

Solution

Question 7

What is the corresponding wing incidence when flying straight and level at a speed of 200 m/s (400 knots approx.) at a height of 1,000 ft?

Note: Neglect change in density 0–1,000 ft *for simplicity.*

Solution

Question 8

What is the maximum normal acceleration that can be achieved at a height of 30,000 ft when flying at a speed of 225 m/s (450 knots)? Neglect compressibility effects and assume adequate thrust is available to counteract the increase in drag and maintain speed of 450 knots at $\alpha_{\max}$.

Solution